

Data Mining Notes

Academic Year 2026/2027

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Chapter 1

Introduction to Data Mining and the KDD Process

1.1 Definition and Fundamental Motivations

The constant technological progress in data generation and storage has led to an exponential growth of available information, both in commercial and scientific contexts. The modern paradigm of data collection is often summarized in the principle of gathering any possible data, anywhere and anytime, in the expectation that such data may acquire value not only for the original purpose for which they were collected, but also for secondary purposes not initially foreseen.

Definition 1.1.1 (Data Mining). **Data Mining** is the application of efficient techniques and algorithms to the analysis of large data collections, aimed at the extraction of valid, novel, potentially useful, and interpretable patterns (hidden knowledge).

Data Mining does not consist in the simple execution of queries on traditional databases (for example, retrieving the average balance via SQL), but rather in the automatic or semi-automatic discovery of non-trivial structures, dependencies, or regularities in the data.

1.1.1 Application Perspectives

The motivations behind the adoption of Data Mining techniques can be distinguished primarily according to two points of view:

- **Commercial point of view:** Companies manage enormous volumes of data (*data warehousing*): web data (Yahoo, Facebook), e-commerce

transactions and retail sales (Amazon), credit card records and bank accounts. In the face of increasingly cheap and accessible computing power, competitive pressure requires extracting value from such masses of data to improve and personalize the services offered (for example in *Customer Relationship Management* - CRM).

- **Scientific and social point of view:** In disciplines such as astronomy (*Sky Survey Data*), Earth observation (NASA EOSDIS satellites collect petabytes of environmental data per year), genomics (gene expression, proteomics), neuroimaging (fMRI), and complex numerical simulations, data are generated at such speeds as to make manual analysis impossible. Data Mining supports scientists in automatic analysis and in the guided formulation of new hypotheses.

Big Data represent to all intents and purposes a proxy of social life (purchasing behaviors, lifestyles, opinions, relationship networks, territorial mobility), offering great opportunities to address global issues: from the efficiency of healthcare to the search for clean energy sources, from forecasting climate impacts to optimizing agricultural production.

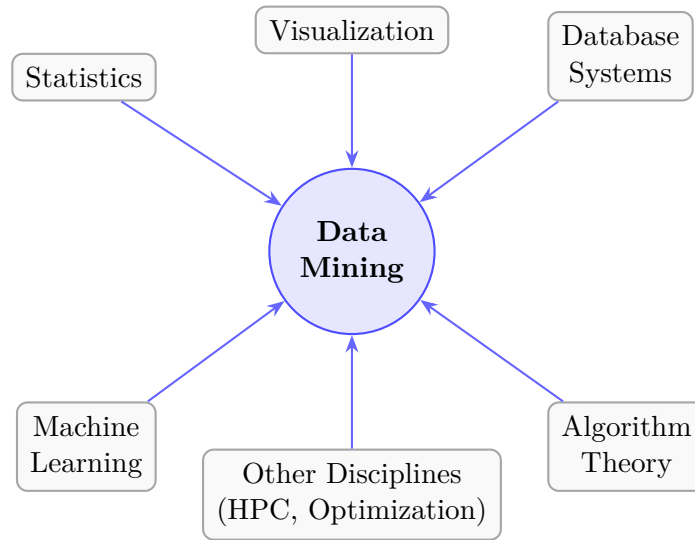
In this modern scenario, applied Artificial Intelligence rests on the foundational paradigm:

$$\mathbf{AI} = \mathbf{Machine\ Learning} + \mathbf{Big\ Data} \quad (1.1)$$

where the availability of massive databases (*Big Data*) provides the empirical substrate and source of information, while *Machine Learning* and *Data Mining* algorithms act as the logical-inductive engine capable of extracting non-trivial generalizations and decision patterns.

1.1.2 A Confluence of Disciplines

Data Mining is not an isolated discipline, but rather a synergistic confluence of several areas of computer science and applied mathematics:



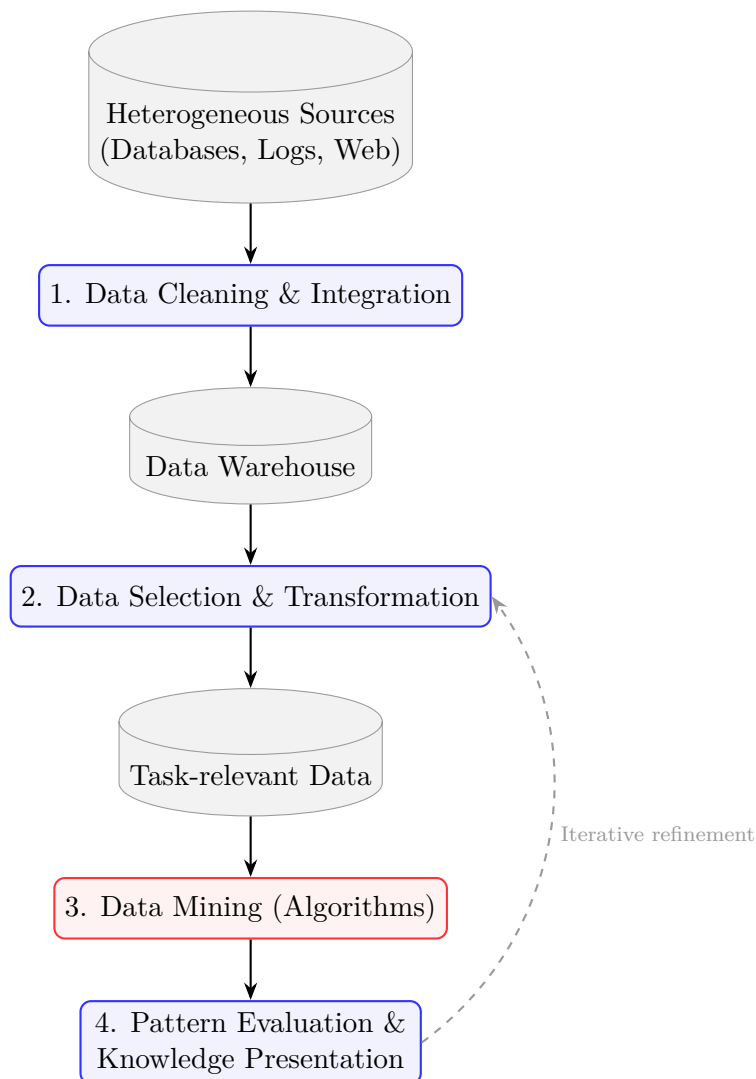
1.2 Types of Data Origin: Primary and Secondary

In setting up an analytical project, it is essential to distinguish the origin of the analyzed data:

- **Primary Data:** Original data collected specifically for a precise investigation objective. They have not undergone any preventive alterations or manipulations by third parties.
- **Secondary Data:** Data that had already been previously collected for operational purposes or different from the current one and made available to the analyst. They often derive from the merging of heterogeneous sources and require deep normalization and quality verification operations.

1.3 The KDD Process (Knowledge Discovery in Databases)

Data Mining constitutes a specific, although central, phase within the broader knowledge discovery process, known as **KDD** (*Knowledge Discovery in Databases*). The KDD is an iterative and interactive process composed of several sequential phases with feedback loops:



Each phase performs specific tasks:

1. **Business Understanding:** Understanding the domain objectives and translating the business problem into a formal data analysis problem.
2. **Data Cleaning and Data Integration:**
 - *Data Cleaning:* removal of noise, correction of syntactic errors, and handling of inconsistencies.
 - *Data Integration:* aggregation and reconciliation of data from multiple and heterogeneous sources, typically flowing into a **Data Warehouse** (a structured and optimized repository for analysis and decision support).

3. Data Selection and Data Transformation:

- *Data Selection*: extraction of the subset of attributes and records strictly relevant to the investigated task (*task-relevant data*).
- *Data Transformation*: transformation of features through aggregations, rescalings, normalizations, or discretizations so that they are suitable for the algorithm.

4. **Data Mining**: The actual execution of intelligent algorithms for the extraction of patterns and mathematical models.

5. **Pattern Evaluation and Knowledge Presentation**: Identification of truly significant patterns through objective and subjective metrics, elimination of redundancy, and graphical visualization of results using tools interpretable by the end user.

1.4 Taxonomy of Data Mining Tasks

Following the classification of Fayyad et al. [1], Data Mining tasks are divided into two macro-categories:

- **Prediction Methods**: Aim to estimate the future or unknown value of a specific target variable starting from the values of other observed variables (predictor attributes).
- **Description Methods**: Aim to identify human-readable and interpretable patterns that describe the intrinsic structure and hidden relationships in the data.

1.4.1 Predictive Tasks: Classification

Classification aims to learn an objective function f that maps each attribute record x into a specific discrete class label $y \in \mathcal{Y}$.

The dataset is divided into two disjoint sets:

- **Training Set**: Used to train the classifier.
- **Test Set**: Used to evaluate the generalization capability of the model on unseen instances during the learning phase.

Example 1.4.1 (Didactic Example: Decision Tree Induction for Credit Worthiness). Consider a standard bank credit lending problem, where the objective is to predict whether a new applicant is financially reliable (*Credit Worthy* $\in \{\text{Yes}, \text{No}\}$).

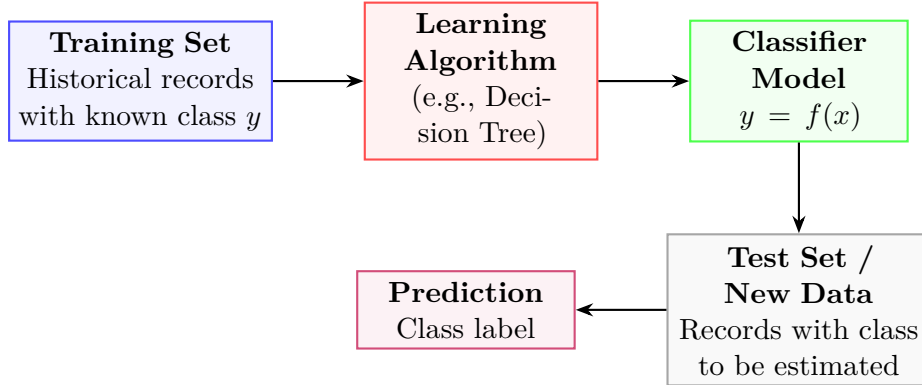


Figure 1.1: Conceptual workflow of the classification process.

1. Training Set (Historical data with known class labels): The lending institution has historical records of past customers with observed loan repayment outcomes:

TID	Employed	Level of Education	# Years at Address	Credit Worthy (Class)
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes

Table 1.1: Training set used to train the credit classification model.

2. Learned Model (Decision Tree): The classification algorithm extracts a hierarchical sequence of logical decisions from the historical data (Fig. 1.2):

3. Test Set (Unseen new instances): When new credit applications arrive with unknown class labels (?), the model evaluates them by traversing the decision tree: This example highlights that the classifier does not

TID	Employed	Education	Years	Target	Model Prediction
5	Yes	Undergrad	7	?	Yes (Employed $\rightarrow$ Undergrad $\rightarrow 7 > 3$)
6	No	Graduate	3	?	No (Unemployed)
7	Yes	High School	2	?	No (Employed $\rightarrow$ High School $\rightarrow 2 \leq 3$)

Table 1.2: Evaluation of the test set using the learned decision tree.

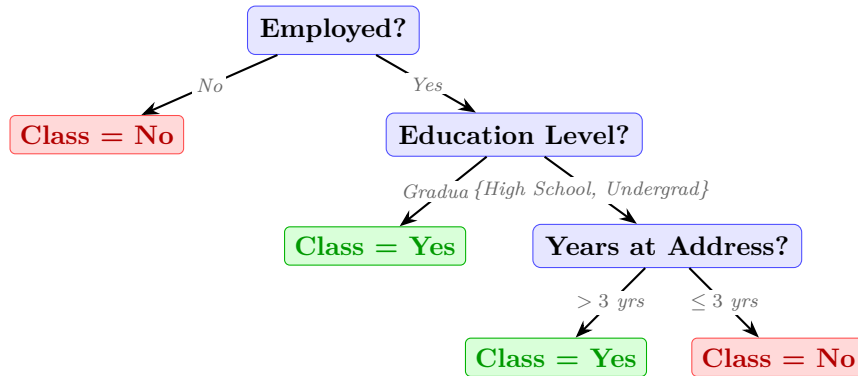


Figure 1.2: Classifier model: learned decision tree.

merely memorize the training instances, but generalizes predictive logical rules applicable to any novel instance.

Example 1.4.2 (Classification Applications). Among the main and most widespread application areas of classification models we find:

- **Credit Worthiness Evaluation:** Predicting whether a loan applicant will repay the debt (*Yes/No*) based on employment, education level, and years of residence.
- **Bank Fraud Detection:** Classifying credit card transactions as legitimate or fraudulent. Features include amount, time, geographical location, and historical spending profile.
- **Churn Prediction:** Estimating the probability that a telecommunications customer will abandon the service for a competitor, analyzing call duration, data traffic volume, and recorded complaints.
- **Other applications:** Automatic categorization of news (sports, finance, politics), clinical diagnosis (benign or malignant tumor), land cover classification from satellite images.

1.4.2 Descriptive Tasks

1. Clustering

Clustering consists in organizing a set of objects into distinct groups (called *clusters*), such that:

1. The distance between objects belonging to the same cluster (**intra-cluster distance**) is **minimized** (maximum internal homogeneity).

2. The distance between objects belonging to different clusters (**inter-cluster distance**) is **maximized** (maximum external separation).

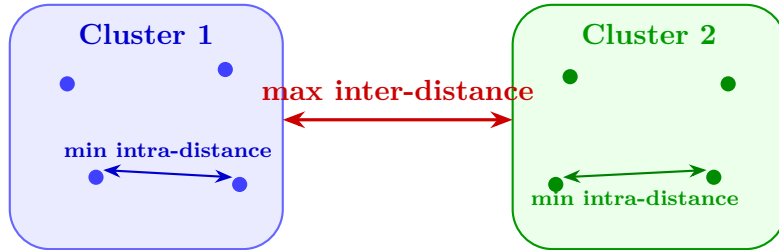


Figure 1.3: Fundamental geometric principle of clustering.

Clustering is an unsupervised technique: it does not require a priori classes.

- **Market Segmentation:** Partitioning customers by demographic traits and purchasing habits to target differentiated marketing campaigns.
- **Document Clustering:** Grouping similar textual documents based on the joint frequencies of keywords (for example, the analysis of the Enron corporate email archive).
- **Summarization and Earth Sciences:** Reducing the complexity of massive spatial series by partitioning sea surface temperature (*Sea Surface Temperature* - SST) and primary production (*Net Primary Production* - NPP) measurements in the Northern and Southern hemispheres.

2. Association Analysis (Discovery of Association Rules)

Given a set of transactions, each consisting of a subset of items taken from a global set of items, association analysis aims to discover probabilistic dependencies in the form of rules:

$$X \implies Y \tag{1.2}$$

where X and Y are disjoint itemsets ($X \cap Y = \emptyset$). The rule indicates that the presence of the items in the antecedent X systematically predicts the contextual presence of the item or items in the consequent Y .

Example 1.4.3 (Market Basket Analysis). Considering a history of cash register receipts: The algorithm extracts rules with high statistical evidence,

TID	Purchased Items
1	Bread, Coca-Cola, Milk
2	Beer, Bread
3	Beer, Coca-Cola, Diapers, Milk
4	Beer, Bread, Diapers, Milk
5	Coca-Cola, Diapers, Milk

such as:

$$\{\text{Milk}\} \implies \{\text{Coca-Cola}\}, \quad \{\text{Diapers, Milk}\} \implies \{\text{Beer}\}$$

Such information guides the strategic placement of goods on shelves and the design of cross-selling offers.

Other applications: Diagnosis of alarms in telecommunication networks (identifying temporal patterns of network alarms that trigger jointly to identify systemic root failures); medical informatics (correlation between combinations of clinical symptoms, blood tests, and emerging pathologies).

3. Anomaly / Deviation / Change Detection

It consists in identifying objects that deviate significantly from the average or expected behavior of the population. Anomalies can be generated by instrumental failures or entry errors (and in this case represent *noise* to be purged), or they can reflect exceptional events of extreme application interest (credit fraud, cyber intrusions against corporate networks, monitoring of sudden deforestation of forest cover via remote sensors).

1.5 Methodological and Computational Challenges in Data Mining

The application of traditional statistical analysis and database methods is often inadequate due to five primary challenges:

1. **Scalability:** Algorithms must handle data volumes on the order of terabytes or petabytes without computational complexity and memory footprint exploding.
2. **High Dimensionality:** The presence of hundreds or thousands of attributes exposes one to the so-called *curse of dimensionality*, where

the geometric space becomes extremely empty and distances tend to lose discriminative power.

3. **Complex and Heterogeneous Data:** Need to process semi-structured or unstructured structures (graphs, texts, genomic sequences, spatiotemporal series).
4. **Data Ownership and Distribution:** Data are frequently geographically distributed across multiple organizations and subject to strict legal privacy and security constraints.
5. **Non-Traditional Analysis:** Need to extract knowledge from continuous streams in real-time (*data stream mining*) without the possibility of storing the entire dataset.

Chapter 2

Data Understanding and Data Exploration

2.1 Objectives of Data Understanding

The data understanding phase (*Data Understanding*) constitutes an indispensable prerequisite for any Data Mining activity. Before selecting or applying any algorithm, the analyst must acquire an overall view of the structure, distribution, and intrinsic quality of the dataset with respect to the project objectives.

In this preliminary phase, it is necessary to answer crucial methodological questions:

- What types of attributes compose the dataset?
- What is the overall level of data quality (presence of noise, errors, or duplicates)?
- How are the numerical and categorical variables distributed?
- Are there evident correlations or redundancies among the attributes?
- How are outliers structured and what application meaning do they assume?
- What strategies should be adopted for missing values?
- Is it necessary to derive or transform attributes to make mining more effective?

2.2 Types of Datasets

The collected data present themselves in different logical-geometric structures based on the nature of the application domain:

1. Record Data:

- **Data Matrix:** A set of objects described by the same fixed set of numerical attributes. A dataset of m objects and n attributes is formally represented by an $m \times n$ matrix:

$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,n} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m,1} & x_{m,2} & \cdots & x_{m,n} \end{pmatrix} \quad (2.1)$$

In this formalization, each object x_i is a point in an n -dimensional vector space $\mathbb{R}^n$.

- **Document Data (Bag-of-Words):** Each textual document is represented by a term vector (*term vector*). Each component of the vector corresponds to a word in the vocabulary, and the associated value indicates its frequency of occurrence in the text.
 - **Transaction Data:** Each record represents a transaction (identified by a TID) and contains a variable set of purchased items or observed events.
2. **Graph Data:** Structures composed of nodes and edges that model relationships and connectivity. Examples include the network of hyper-text links of the World Wide Web, social network graphs, and molecular structures (for example, the planar molecule of benzene C_6H_6).

3. Ordered Data:

- **Sequences of transactions:** Temporal successions of purchases or events for each user.
- **Genomic data:** Discrete biological sequences composed of nucleotide characters $\{\text{A}, \text{C}, \text{G}, \text{T}\}$.
- **Spatio-Temporal Data:** Continuous or sampled measurements over geographical coordinates and regular time intervals (for example, the monthly average temperature of the land and ocean surface).

2.3 Objects and Attributes

A dataset is a collection of **data objects** and their corresponding **attributes**.

- An **object** (also known as a record, point, case, sample, entity, or instance) represents a single entity of the domain of interest.
- An **attribute** (also known as a variable, field, characteristic, dimension, or feature) represents a measurable property or characteristic of an object (for example, age, taxable income, or eye color).

2.3.1 Measurement Scales of Attributes

The type of measurement scale of an attribute is defined by the set of algebraic properties and mathematical operators that preserve the meaning of the measurement:

1. **Distinctness** ($=, \neq$): Possibility to distinguish whether two values are equal or different.
2. **Order** ($<, >$): Possibility to establish an order relationship between values.
3. **Meaningful differences** ($+, -$): Presence of a constant unit of measurement that allows quantifying the interval between two values (zero is conventional/arbitrary).
4. **Meaningful ratios** ($*, /$): Presence of an **intrinsic absolute zero** (indicating the total absence of the measured quantity), which makes the proportional comparison between quantities meaningful.

2.3.2 Further Distinctions of Attributes

- **Binary Attributes:** Nominal attributes that exclusively admit two states (0 and 1). They are distinguished into:
 - *Symmetric Binary:* Both outcomes have equal informational importance (for example, biological gender).
 - *Asymmetric Binary:* One of the two outcomes is rare but of extreme informational relevance (for example, a positive outcome to a diagnostic test for a rare disease). By convention, the value 1 is associated with the event of greater interest.

Scale	Properties	Description	Examples	Statistical Operators
Nominal	Distinctness ($=, \neq$)	Qualitative values or categorical labels without intrinsic ordering	SSN, ZIP code, eye color, gender	Mode, entropy, χ^2 test, contingency
Ordinal	Distinctness and Order ($=, \neq, <, >$)	Values with meaningful ordering but non-constant interval	Educational degree, Likert scale (1–10), mineral hardness	Median, percentiles, rank correlation (Spearman)
Interval	Order and Differences ($+, -$)	Quantitative differences with meaning; zero is conventional	Temperature in $^{\circ}\text{C}$ or $^{\circ}\text{F}$, calendar dates	Mean, variance, standard deviation, Pearson, t and F tests
Ratio	All 4 properties ($=, <, +, *, /$)	Presence of absolute zero; ratios between physically valid measurements	Age, income, mass, counts, Kelvin	Geometric and harmonic mean, percentage variation

Table 2.1: Taxonomy of attribute measurement scales (Stevens).

- **Discrete vs Continuous Attributes:**

- *Discrete*: Assume a finite or countably infinite set of values (typically represented by integers, counts, or categories). Binary attributes constitute a special case of discrete attribute.
- *Continuous*: Assume values in a continuous subset of real numbers $\mathbb{R}$ (temperature, weight, coordinates). In computational practice, they are measured and stored using floating-point representations (*floating-point*) with finite precision.

2.4 Data Quality

Data quality is a critical factor for the success of any analytical initiative. As highlighted in Thomas Redman's studies, problems arising from poor quality data typically cost organizations between 10% and 20% of their overall revenues. If a credit risk model is trained on flawed or erroneous data, the system will reject solvent debtors and disburse loans to insolvent individuals.

2.4.1 Dimensions of Data Quality

Quality problems manifest themselves under various dimensions:

1. **Syntactic Accuracy:** The recorded value violates formal constraints or does not belong to the predetermined domain (for example, the string "fmale" for gender, or the presence of letters in a numeric field). Such anomalies can be identified immediately and automatically through simple schema and type checks.
2. **Semantic Accuracy:** The data is formally correct and conforms to the expected type, but false in the reference reality (for example, associating the name "John Smith" with the female gender or recording a biological age of 180 years). Detecting these discrepancies requires domain rules (*business rules*) or cross-referencing with certified external sources.
3. **Completeness:** Systematic lack of values for certain attributes or absence of entire records. An insidious violation of completeness is **sampling bias** (for example, the analysis of insolvency risk carried out on a historical archive that only includes customers already approved by the bank, excluding a priori rejected low-income profiles).

4. **Unbalanced Data:** Strongly asymmetric distribution of classes (for example, fraudulent transactions equal to 0.1% of the total or defective products on highly automated production lines).
5. **Timeliness:** Obsolete degree of data updating compared to the current dynamics of the observed phenomenon.
6. **Duplicate Data:** Presence of identical or nearly identical records relating to the same entity (for example, the same customer registered multiple times with different email addresses following the merger of two heterogeneous databases). Such issues require dedicated deduplication and *record linkage* processes.

Anomalies in Measurement Streams and Sensors

In industrial monitoring and IoT systems, a broken or blocked sensor typically transmits a flat, constant value of zero. During the exploratory phase, such readings must not be treated as genuine physical measurements, but rather interpreted and handled in all respects as **missing values** (caused by hardware malfunction).

2.5 Descriptive Statistics and Univariate Exploration

2.5.1 Measures of Central Tendency

- **Sample Mean ($\bar{x}$):** For a sample of size m :

$$\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i \quad (2.2)$$

The mean is a distributive measure (it can be calculated by partitioning the dataset into subsets), but it is **extremely sensitive to the presence of outliers**.

- **Median:** The value that occupies the central position in the increasing order of the observations (or the mean of the two central values if m is even). It is a robust measure, with a breakdown point (*breakdown point*) of 50%.
- **Trimmed Mean:** Mean calculated by excluding a fixed percentage (for example 5% or 10%) of the most extreme values at both ends.

- **Mode:** The value or category that occurs with the maximum absolute frequency. A distribution can be unimodal, bimodal, multimodal, or modeless if all observations appear only once.

2.5.2 Skewness and Multimodality

- **Positively Skewed / Right-skewed:** The right tail of the distribution is elongated. Most observations are concentrated on values lower than the arithmetic mean (Mode < Median < Mean). A typical example is the distribution of **incomes**: the vast majority of citizens earn low or medium incomes, while a small number of individuals earn extremely high incomes, pulling the mean upwards.
- **Bimodal Distribution:** Presence of two distinct peaks in the frequency density. It typically signals the influence of an unstated binary latent variable (for example, the distribution of human body weight in a mixed population, influenced by the binary variable sex).

2.5.3 Measures of Dispersion

- **Range:** Difference between maximum value and minimum value: $\text{Range} = x_{\max} - x_{\min}$.
- **Sample Variance (s^2) and Standard Deviation (s):**

$$s^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2, \quad s = \sqrt{s^2} \quad (2.3)$$

The standard deviation measures the dispersion around the mean; it is zero only if all observations are identical ($s = 0$), otherwise $s > 0$. It is also sensitive to outliers.

- **Robust Measures of Dispersion:**

– *Absolute Average Deviation* (AAD):

$$\text{AAD} = \frac{1}{m} \sum_{i=1}^m |x_i - \bar{x}| \quad (2.4)$$

– *Median Absolute Deviation* (MAD):

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|) \quad (2.5)$$

– *Interquartile Range* (IQR): Distance between the third quartile Q_3 (75th percentile) and the first quartile Q_1 (25th percentile):

$$\text{IQR} = Q_3 - Q_1 \quad (2.6)$$

2.5.4 Box Plot and Five-Number Summary

The **Box Plot** (box and whisker plot) offers a non-parametric visual summary of the distribution based on the **five-number summary**:

$$[\text{Minimum}, Q_1, \text{Median}(Q_2), Q_3, \text{Maximum}]$$

Its geometric structure relies on the concept of quantile: given $p \in (0, 100)$, the $p\%$ -**quantile** (or p^{th} percentile) is the cutoff value x_p such that $p\%$ of sample observations are smaller than or equal to x_p , while $(100 - p)\%$ are greater. Specifically:

- Q_1 corresponds to the 25%-quantile (first quartile).
- The median Q_2 corresponds to the 50%-quantile (second quartile).
- Q_3 corresponds to the 75%-quantile (third quartile).
- The width of the central box represents the interquartile range $\text{IQR} = Q_3 - Q_1$, encompassing the central 50% of the data mass.

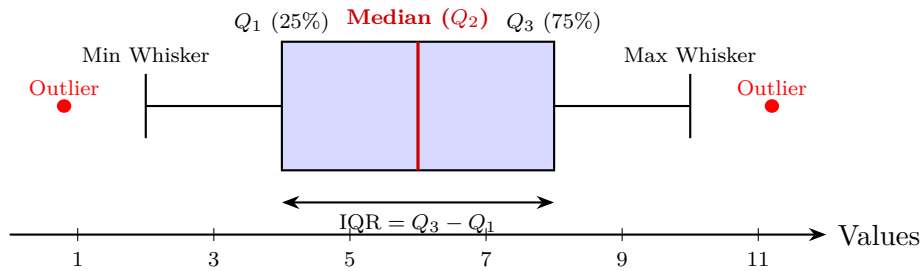


Figure 2.1: Geometric anatomy of a Box Plot with identification of outliers.

The limits of the whiskers are conventionally calculated by extending up to the most extreme observed value within the interval $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$. The observations that fall outside such thresholds are visually labeled as **outliers**.

A key extension for bivariate and multivariate exploration is given by **Conditional Box Plots**: the distribution of a continuous numerical attribute is stratified and displayed by placing box plots side by side for each category of a discrete attribute (typically the target class or a natural clustering). Figure 2.2 illustrates the distribution of *Petal Length* stratified across the three flower species of the Iris dataset, highlighting that *Setosa* is distinctly separable with very low dispersion, whereas *Versicolor* and *Virginica* exhibit a slight overlap in their interquartile ranges.

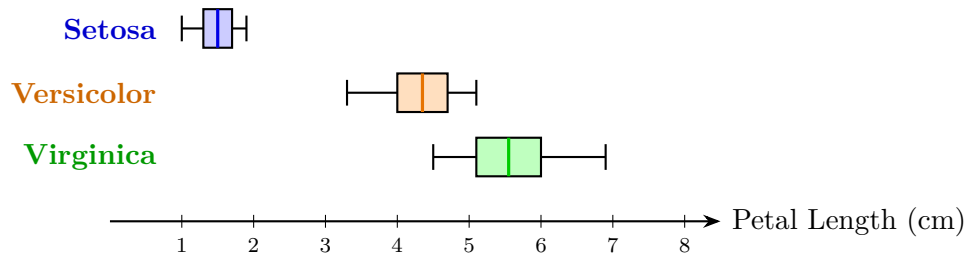


Figure 2.2: Example of Conditional Box Plot: distribution of the *Petal Length* attribute conditioned on the flower species in the Iris dataset.

Critical Comparison: Box Plot vs Histogram

Although the box plot effectively synthesizes position and dispersion, **it is not able to show the real shape of the distribution**. Two distributions having radically different densities (for example, a uniform unimodal distribution and a strongly bimodal distribution with an empty central valley) can produce identical values of Q_1 , median, Q_3 , Minimum, and Maximum, and therefore generate the exact same box plot. The histogram is essential to detect multimodality.

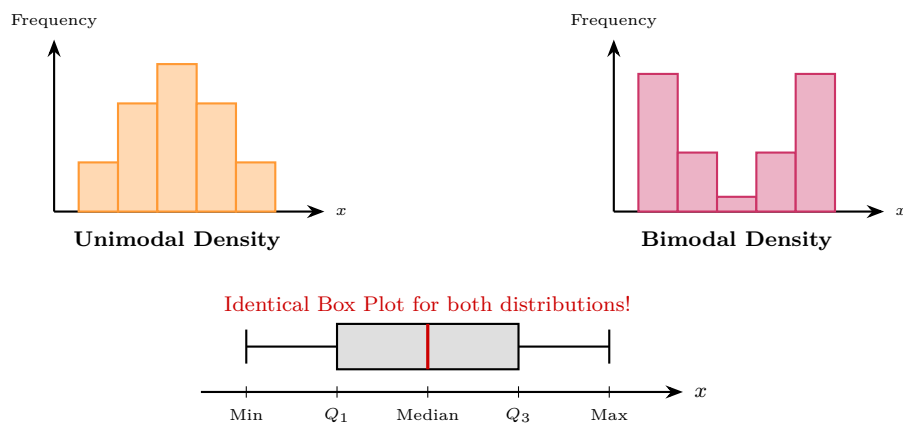


Figure 2.3: Critical comparison: two distributions with diametrically opposite behaviors (unimodal vs bimodal with empty center) generate the same five-number summary and the same Box Plot.

2.5.5 Bar Chart for Categorical Attributes

The **Bar Chart** visually represents the frequency distribution of a qualitative or categorical attribute. Each discrete category is represented by a vertical (or horizontal) bar, whose height corresponds to the absolute count or relative frequency. Unlike histograms, bars are separated by empty spaces to highlight that the modalities do not constitute a numerical continuum.

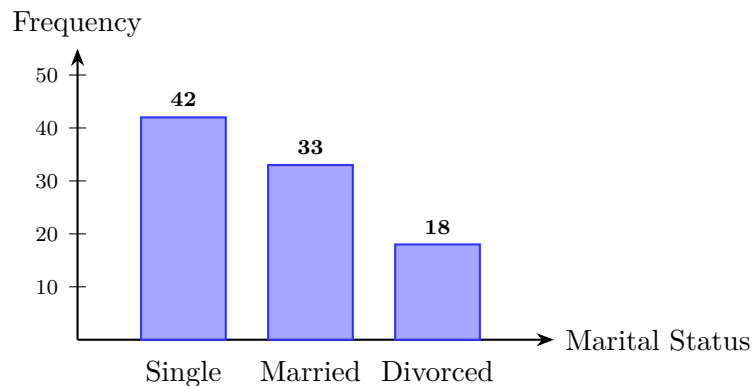


Figure 2.4: Example of a Bar Chart for a nominal categorical attribute (Marital Status).

2.5.6 Histograms for Numerical Attributes and Choice of Bins

A **histogram** shows the empirical frequency distribution of a continuous numerical attribute. The domain of variability of the quantity is partitioned (*discretized*) into a predetermined number k of contiguous intervals called *bins* (or class widths). For each bin, the height of the overlying bar is proportional to the number of sample observations that fall within it.

An incorrect choice of the number of bins profoundly alters the perception of the data:

- An insufficient number of bins causes *oversmoothing*, leveling the distribution and masking skewnesses and multimodality.
- An excessive number of bins causes *undersmoothing*, fragmenting the sample into discontinuous frequencies and emphasizing sampling noise.

For approximately normal unimodal distributions and moderate sample sizes n , the standard estimate for the optimal number of intervals is given by

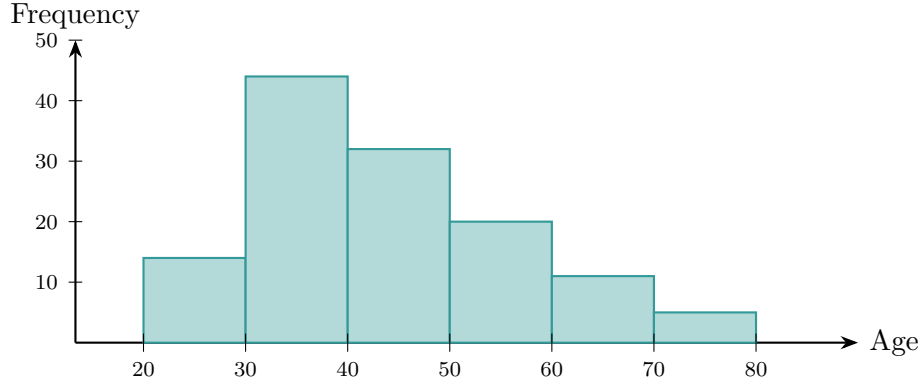


Figure 2.5: Example of a Histogram for a continuous numerical attribute with contiguous bins of constant width.

Sturges' rule:

$$k = 1 + \lceil \log_2(n) \rceil = 1 + \lceil 3.322 \log_{10}(n) \rceil \quad (2.7)$$

where n represents the sample cardinality. This heuristic explicitly assumes a Gaussian-like distribution. When the sample size grows considerably or the underlying distribution exhibits multiple modes or heavy skewness, Sturges' rule tends to generate an insufficient number of bins; in such cases, it is crucial to test alternative values of k experimentally (e.g., comparing low-, medium-, and high-resolution partitionings such as 5, 17, or 50 bins) to assess the robustness of observed patterns.

2.6 Multivariate Visualization Techniques

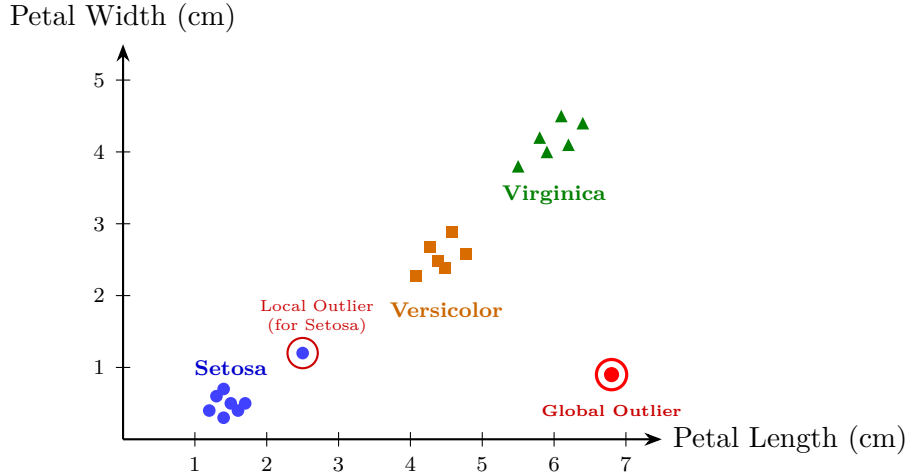
2.6.1 Scatter Plot and Multi-Dimensional Visualization

The **Scatter Plot** treats pairs of numerical attributes as Cartesian coordinates (x_i, y_i) in the plane. It is the main visual tool for:

- Identifying linear correlations or non-linear functional trends.
- Detecting natural clusters of points (*clusters*).
- Finding multivariate outliers not visible in univariate projections.

Example 2.6.1 (Analysis of the Anderson/Fisher Iris Dataset). The famous *Iris* dataset (150 samples of flowers belonging to three species: *Setosa*, *Versicolor*, *Virginica*) describes 4 numerical attributes: sepal length and width,

petal length and width. The bivariate scatter plot shows that the attributes related to the petal (*Petal Length* vs *Petal Width*) guarantee a clear separation among the species, also allowing the isolation of potential anomalies.



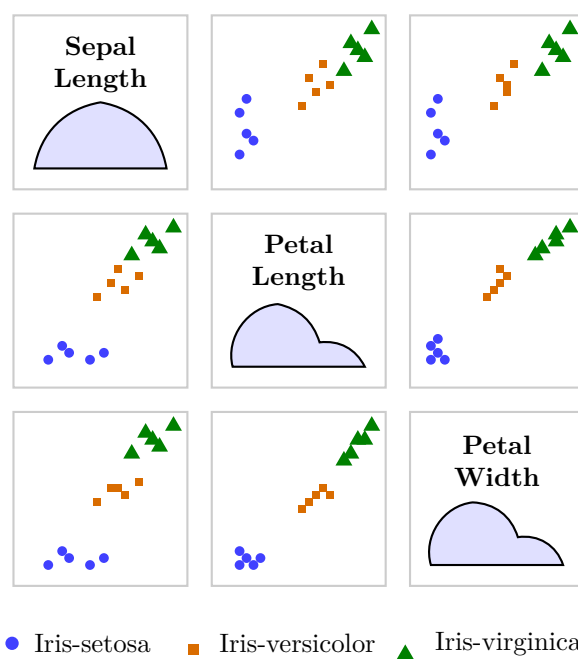


Figure 2.7: Scatter Plot Matrix (SPLOM) of the Iris dataset with the pairwise comparison for all three species and the univariate density estimations along the diagonal.

2.6.3 High-Dimensionality Visualization

Parallel Coordinates

When the number of dimensions exceeds 3, the conventional Cartesian representation becomes inapplicable. **Parallel Coordinates** replace the orthogonal axes with a set of equidistant vertical parallel axes, each corresponding to an attribute.

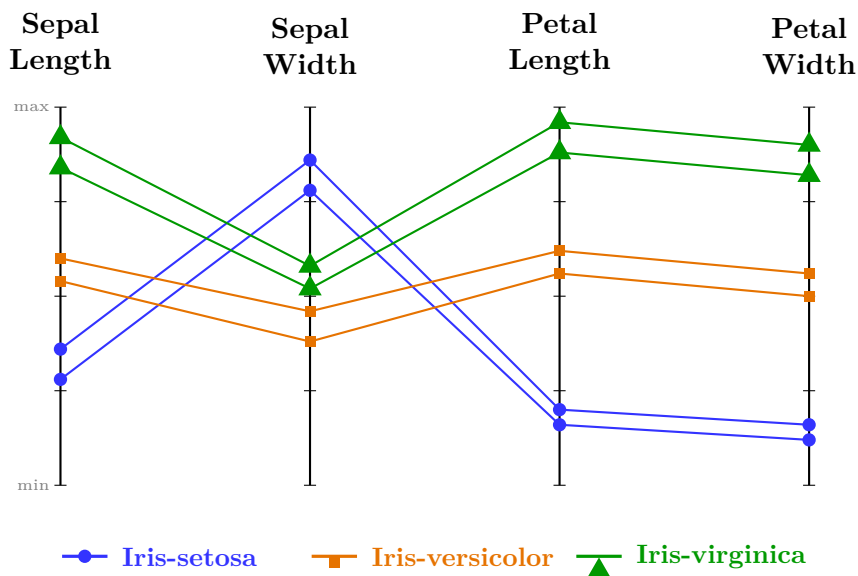


Figure 2.8: Representation of 4-dimensional data using Parallel Coordinates for all three species of the Iris dataset.

- Each multidimensional object is represented by a **polygonal broken line** that connects the respective scaled values along the vertical axes.
- Objects belonging to the same class tend to form bundles of parallel lines or with coordinated trends.
- **Sensitivity to the ordering of the axes:** The visibility of clusters and correlations depends critically on the sequential arrangement of the axes. Two strongly correlated attributes show regular geometric patterns only if positioned as adjacent axes.

Radar Plot and Star Plot

In **Radar Plots**, the axes start from a common center and radiate out in a star shape at constant angular intervals. The value of each attribute defines

a vertex along its own radius; joining the vertices yields a polygon for each instance. **Star Plots** apply the same principle by drawing each object in a separate graph, facilitating comparison for individual profiles.

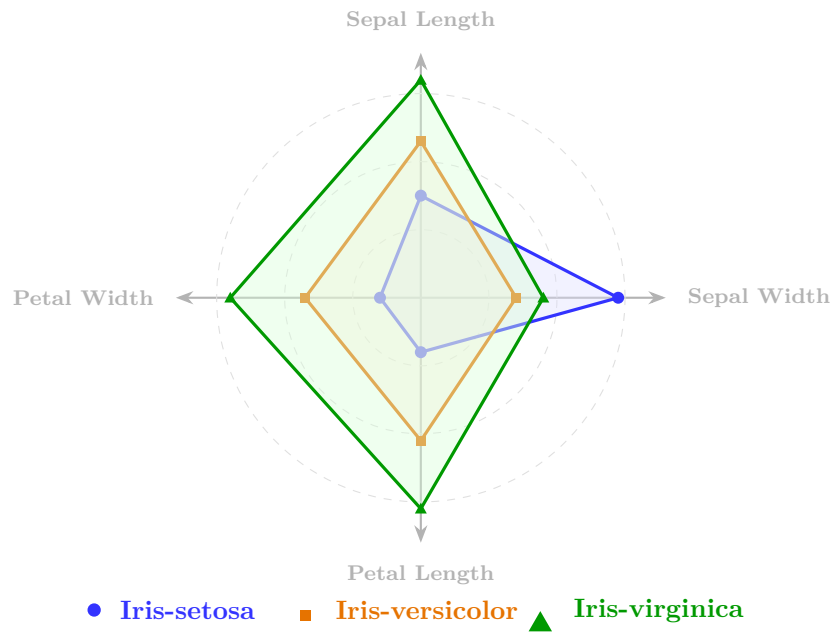


Figure 2.9: Example of a Radar Plot with visual comparison of multivariate profiles for all three Iris species.

2.7 Linear Correlation and Association Analysis

Correlation analysis aims to identify, quantify, and interpret the strength and direction of the linear relationship between pairs of continuous numerical attributes.

2.7.1 Motivations and Multicollinearity in Predictive Models

Within the data understanding workflow, assessing correlation between variables serves several methodological goals:

- **Identifying informative redundancies:** If two attributes exhibit nearly identical trajectories, one of them can be discarded without

appreciable loss of descriptive power, reducing the dimensionality of the feature space.

- **Preparation for predictive models and clustering:** Many machine learning algorithms (such as linear regression, logistic regression, or distance-based clustering) assume independent features or benefit substantially from the removal of collinear attributes.
- **Detecting multicollinearity:** Occurs when two or more predictor variables (*features*) in a regression model are strongly linearly related to one another. Consider the paradigmatic example of predicting house prices using simultaneously *surface area in square metres* (m^2) and *surface area in square feet* (ft^2). Both variables convey exactly the same information up to a constant conversion factor. Consequently, the regression model experiences severe numerical instability, struggling to disentangle the individual marginal contributions of each predictor and inflating estimator variance.

2.7.2 From Variance to Sample Covariance

While sample variance quantifies the dispersion of observations for a single variable around its own mean:

$$s_x^2 = \text{Var}(X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (2.8)$$

in data mining applications interest is constantly focused on pairwise interactions between attributes:

- Do customers with higher taxable incomes spend larger sums?
- Does a product's unit price lead to a contraction in sales volume?
- Do students who study more weekly hours systematically attain higher exam grades?
- Do two physical sensors arranged in parallel provide largely redundant measurement streams?

To quantify how two numerical quantities vary jointly, we introduce **sample covariance**:

Definition 2.7.1 (Sample Covariance). Given a sample of n observation pairs (x_i, y_i) , the sample covariance between X and Y is defined as:

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad (2.9)$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ represent the respective sample means.

A fundamental theoretical property dictates that variance is simply the covariance of a variable with itself:

$$\text{Cov}(X, X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \text{Var}(X) = s_x^2 \quad (2.10)$$

2.7.3 Sign Origin and Geometric Quadrant Interpretation

The algebraic sign of each term in the summation stems from the product of paired deviations from their respective means:

- **Positive concordant deviations:** $(x_i - \bar{x}) > 0$ and $(y_i - \bar{y}) > 0 \implies (+) \times (+) = +$.
- **Negative concordant deviations:** $(x_i - \bar{x}) < 0$ and $(y_i - \bar{y}) < 0 \implies (-) \times (-) = +$.
- **Discordant deviations:** $(+) \times (-) = -$ or $(-) \times (+) = -$.

Translating the Cartesian origin to the sample centroid $(\bar{x}, \bar{y})$, the two-dimensional space is divided into four quadrants (Figure 2.10):

1. If observations in quadrants **I and III** (concordant deviations) dominate, positive products prevail and the sample covariance is strictly positive ($\text{Cov}(X, Y) > 0$).
2. If quadrants **II and IV** (discordant deviations) prevail, negative products dominate and the covariance is strictly negative ($\text{Cov}(X, Y) < 0$).
3. If observations are uniformly or symmetrically scattered across all four quadrants, positive and negative terms cancel each other out, resulting in a covariance close to zero ($\text{Cov}(X, Y) \approx 0$).

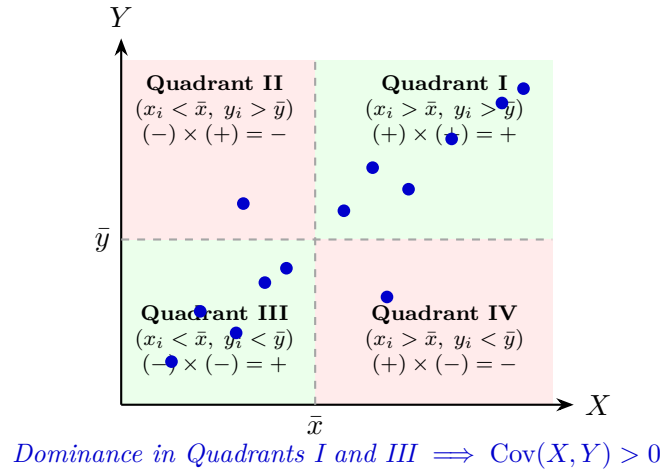


Figure 2.10: Geometric interpretation of covariance across quadrants centered at $(\bar{x}, \bar{y})$. Observations in concordant quadrants I and III yield positive contributions, whereas discordant quadrants II and IV yield negative contributions.

2.7.4 Limitation of Covariance: Scale and Unit Dependence

Although covariance unambiguously reveals the direction of linear agreement, it suffers from a critical drawback: **it depends strictly on the measurement units of the variables**. Consider the relationship between annual income and household expenditure:

- If both income and expenditure are measured in euros, the unit of $\text{Cov}(\text{Income}, \text{Expenditure})$ is euro².
- If income is converted into *thousands of euros*, the numerical covariance drops by a factor of 1000, even though the substantive economic relationship between both quantities remains strictly unchanged.

Consequently, covariance indicates whether the linear trend is positive or negative, but its absolute magnitude cannot be compared across different scales or attribute pairs without standardization.

2.7.5 Pearson Linear Correlation Coefficient

To obtain a unit-free, standardized metric, sample covariance is divided by the product of the respective sample standard deviations:

Definition 2.7.2 (Pearson Correlation Coefficient). Given a sample of n pairs (x_i, y_i) , the sample linear correlation coefficient r_{xy} is defined as:

$$r_{xy} = \frac{\text{Cov}(X, Y)}{s_x s_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (2.11)$$

where s_x, s_y are the sample standard deviations of X and Y .

Core properties of the Pearson correlation coefficient:

- **Dimensionless and scale-invariant:** r_{xy} is a pure number, invariant under any positive affine transformation ($X' = aX + b$, with $a > 0$).
- **Bounded range:** $-1 \leq r_{xy} \leq +1$.
- **Perfect collinearity** ($|r_{xy}| = 1$): Points lie precisely on a straight line; with positive slope if $r_{xy} = +1$, and negative slope if $r_{xy} = -1$.
- **Absence of linear association** ($r_{xy} \approx 0$): Signals little to no linear relationship.
- **Blindness to non-linear dependencies:** As illustrated in Figure 2.11(d), zero correlation ($r_{xy} \approx 0$) does not imply statistical independence: deterministic quadratic or sinusoidal relationships may exhibit exactly zero Pearson correlation.

2.7.6 Statistical Association vs Causation and Confounding Factors

A fundamental tenet in data science is the epistemological boundary separating association from causation:

$$\text{Correlation} \neq \text{Causation} \quad (2.12)$$

Consider an academic dataset where weekly study hours and examination marks exhibit a high positive correlation ($r_{xy} = +0.98$). Students with fewer study hours tend to score below the mean, while those studying longer hours score above it. Although this demonstrates an undeniably strong positive linear association, **it does not warrant the conclusion that 2 additional study hours will deterministically cause an increase in marks**. Multiple confounding variables (such as prior academic background, lecture attendance, or classroom attention) may influence both quantities simultaneously.

Linear correlation:

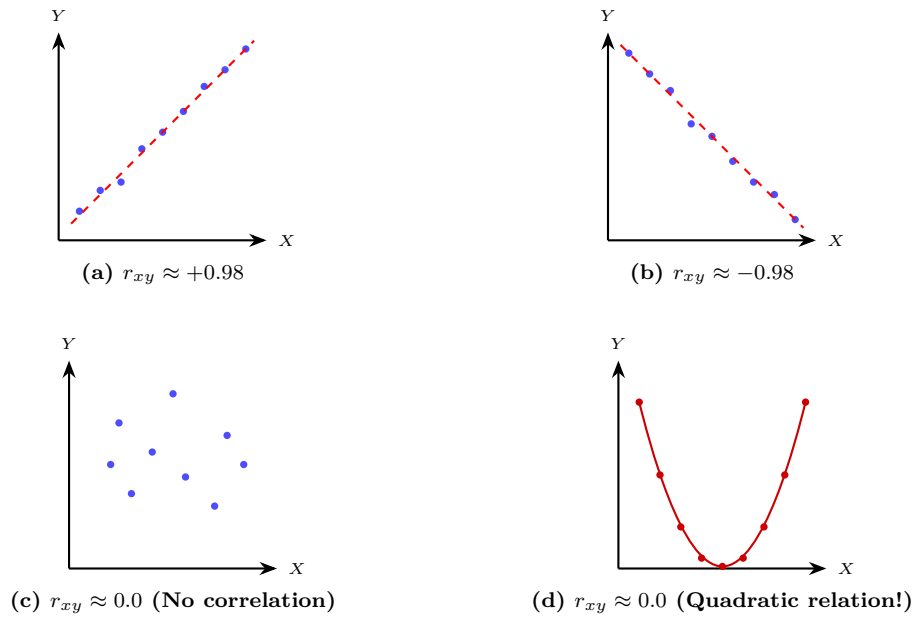


Figure 2.11: Visual interpretation of the Pearson correlation coefficient: (a) positive correlation; (b) negative correlation; (c) absence of linear relation; (d) non-linear deterministic dependence missed by Pearson.

1. Does not identify the causal direction ($X \rightarrow Y$ or $Y \rightarrow X$).
2. Can be driven by a **confounding variable** (omitted variable bias).

Example 2.7.1 (The Confounding Variable: Ice Cream and Drowning). In seasonal regional records, ice cream sales and drowning incidents show a strong positive correlation. Eating ice cream clearly does not cause drowning. The observed relationship is driven entirely by an omitted confounding variable: **high summer temperatures** (heatwaves), which simultaneously increase demand for chilled desserts and the number of people swimming, producing a spurious association.

2.7.7 Spearman's Rank Correlation for Monotonic Relationships

When the relationship between two variables is monotonic but non-linear, or in the presence of severe outliers that distort means and standard deviations, the **Spearman rank correlation coefficient** (ρ_s) is preferred.

Spearman does not evaluate original continuous values directly, but replaces each value with its **rank** in ascending order:

Definition 2.7.3 (Spearman’s Rank Correlation Coefficient). Let $\text{rank}(x_i)$ and $\text{rank}(y_i)$ denote the ranks of x_i and y_i . If there are no ties, the coefficient is given by:

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}, \quad \text{with } d_i = \text{rank}(x_i) - \text{rank}(y_i) \quad (2.13)$$

In the presence of tied ranks, ρ_s is formally equivalent to Pearson’s correlation computed directly on the rank vectors.

Essential properties:

- $\rho_s = +1$ when the rankings of both variables are in exactly the same order (**monotonically increasing** relationship).
- $\rho_s = -1$ when the rankings are in reverse order (**monotonically decreasing** relationship).
- Measures whether variables tend to move in the same or opposite direction regardless of linearity (e.g., for $Y = X^3$ or $Y = \log(X)$, $\rho_s = 1$).
- Highly robust against isolated extreme outliers.

2.7.8 Correlation Matrix and Feature Selection

In datasets with multiple continuous attributes, all pairwise bivariate correlations are systematically arranged in a symmetric $p \times p$ **correlation matrix**. In the Iris dataset, for example, the matrix reveals an exceptionally strong correlation between *Petal Length* and *Petal Width* ($r \approx 0.96$), signaling feature redundancy and suggesting that dimensionality reduction or feature selection can discard one of the two without sacrificing model performance.

2.8 Handling Outliers and Missing Values

2.8.1 Outliers: Noise vs Analytical Target

An **outlier** is a data object whose characteristics deviate considerably from the vast majority of observations in the dataset:

- **Outliers as noise:** Originating from typographic mistakes, sensor failures, or protocol errors. Such instances severely corrupt sample means, variances, and regression fits, and must be detected and removed (or imputed) during data cleaning.

- **Outliers as the primary analysis target:** Genuine unusual situations that constitute the central analytical goal. Paradigmatic examples include **credit card fraud detection**, **network intrusion detection**, and early diagnosis of **rare medical pathologies**.

Even when outliers represent valid measurements of rare events, it is often advisable to exclude them temporarily when computing summary metrics aimed at describing the typical population.

2.8.2 Outlier Detection Methodologies

Anomalies can be discovered across different dimensional settings:

- **Single Attribute (Univariate Analysis):**
 - *Categorical attributes:* An outlier is a category that occurs with a frequency vastly lower than all other domain classes.
 - *Numerical attributes:* Visual and analytical inspection using Box Plots (points beyond $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$) and standardized z -scores ($z = \frac{x-\mu}{\sigma}$).
- **Multiple Attributes (Multivariate Analysis):**
 - *2D and 3D Scatter Plots:* Reveal both global outliers (isolated with respect to the entire point cloud) and local outliers (deviating from their specific class cluster, as seen in Figure 2.6).
 - *PCA Projections:* Project high-dimensional data onto maximal-variance planes (Principal Component Analysis) to visually isolate anomalies.
 - *Clustering Algorithms:* In density-based or partition algorithms (such as DBSCAN or k-means with distance cutoffs), outliers naturally emerge as points that cannot be assigned to any major cluster or that form isolated micro-clusters.

2.8.3 Missing Values and Camouflage

The absence of values for attributes across records arises from several root causes:

- **Information not collected or withheld:** Survey respondents explicitly refuse to disclose sensitive personal data (e.g., income, weight, or age).

- **Logical non-applicability:** The attribute has no logical meaning for a subpopulation (e.g., annual employment income for school-age children, or driver's licenses for minors).
- **Sensor breakdown or network loss:** Physical hardware malfunctions, depleted sensor batteries, or packet dropouts.

Pitfall of Camouflaged Missing Values: Zeros and Default Sentinels

Frequently, a missing value is not explicitly stored as NULL or NaN, but rather recorded using conventional default or sentinel numbers (such as 0, -1, or 999). A blocked or broken sensor, for instance, often transmits a constant stream of 0s. Unless intercepted during Data Understanding, analytics software treats these sentinel numbers as true physical measurements, severely biasing means, standard deviations, correlations, and classification models.

2.9 Operational Checklist for Data Understanding

Upon concluding the exploration stage, the analyst must have methodically executed the following operational verifications:

1. **Assess Overall Data Quality:** Audit syntactic accuracy (out-of-domain entries) and semantic consistency, check completeness, evaluate timeliness, deduplicate records, and quantify class imbalance.
2. **Detect and Analyze Outliers:** Identify univariate and multivariate outliers using visualization tools (box plots, scatter plots, PCA, parallel coordinates) and determine appropriate handling policies (noise exclusion vs target investigation).
3. **Uncover and Examine Missing Values:** Quantify null frequencies and expose camouflaged missing values hidden behind sentinel zeros or default settings.
4. **Discover and Validate Correlations and Dependencies:** Compute linear (Pearson) and monotonic rank (Spearman) correlation matrices to confirm domain hypotheses, discover unexpected linkages, and prevent multicollinearity and feature redundancy.

5. **Verify Domain-Specific Assumptions:** Test distributional premises required by downstream models or heuristics (e.g., verifying whether an attribute follows an approximately normal distribution before applying Sturges' rule or parametric tests).
6. **Compare Statistics with Expected Behavior:** Ensure that empirical means, medians, skewness, and spreads align consistently with domain expertise and business context expectations.

Bibliography

- [1] Usama Fayyad, Gregory Piatetsky-Shapiro, and Padhraic Smyth. From data mining to knowledge discovery in databases. *AI Magazine*, 17(3):37–54, 1996.